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Utilizing VLM for extracting photos of named individuals from historical newspapers

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GitHub: <https://github.com/abs3nce/knn-project>

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1 Introduction

Extracting structured information from scanned historical newspapers remains a significant challenge due to paper degradation, complex layouts, and low-contrast imagery. Current state-of-the-art object detectors often fail to generalize under these conditions, achieving low accuracy on historical datasets [2]. This project focuses on extracting photographs of people from a single newspaper page by their name. By leveraging Vision-Large-Language Models (VLMs), the goal is to evaluate and surpass a zero-shot baseline via few-shot learning and fine-tuning.

1.1 Related work

Archival Faces: Detection of Faces in Digitized Historical Documents. To address the fact that standard face detectors achieve only around 24% mAP on scanned historical documents, this work introduces a new dataset designed to retrain existing detectors to modern performance standards [2]. While this work focuses on object detection using a standard YOLO model, our project uses VLMs to extract a person’s photo from the surrounding text by their name.

1.2 Dataset

PeopleGator. This dataset comprises approximately 2,500 newspaper pages featuring annotated bounding boxes for figures, bounding polygons for text regions, and corresponding OCR transcriptions.

1.3 Proposed approach and experiments

We propose three experiments (ranging from general inference to domain-specific fine-tuning) to measure their expected incremental performance gains:

1. **Zero-Shot Baseline.** First, we establish a performance baseline using unmodified VLMs. Models will be tasked with finding portrait photographs without prior exposure to historical newspaper pages.
2. **Few-Shot Learning.** Building on the baseline, the next experiment uses few-shot learning. We will provide the model with three annotated example pairs, each containing a newspaper scan with the desired person’s name and the corresponding bounding box coordinates for the person’s portrait.
3. **Fine-Tuning.** The final experiment involves fine-tuning of the VLM using LoRA [1]. This experiment aims to adapt the model’s weights to the unique newspaper features that are non-existent in current printing technologies.

Performance is quantified using Intersection over Union (IoU) for bounding-box accuracy and the F1-score metric for model comparison. This approach will allow us to analyze the pros and cons of lightweight prompt-oriented engineering versus computation-heavy fine-tuning.

2 Models description

The next three subsections will discuss the chosen visual language models and the reasons behind those choices, the inference strategy, and the proposed model output with the annotation results.

2.1 Supported models and their objective

We have selected three models from the same Qwen2.5-VL model family, specifically the 3B¹, 7B² and the 32B³ model sizes to experiment on. On the Hugging Face platform⁴, these models are said to be capable of analyzing text, charts, graphics, and layouts. This is ideal, as this work's main problem is whether VLMs can identify and localize (by drawing bounding boxes) people in historical newspaper scans. It is also reported that these models are proficient in object localization, generating bounding boxes, and outputting structured JSON data.

We have started with the Qwen-3B. Although this model is small, the motivation is to see whether small models, which are easier and cheaper to fine-tune, can succeed on this complex task.

To determine how the model size affects the results, we included the Qwen-7B model to see whether a roughly twice-as-large model could achieve substantially better accuracy. Finally, the Qwen-32B was selected to test the performance of a much larger, highly capable model. The primary purpose of this selection is to compare performance across different scales and determine whether scaling up the parameters yields significant improvements or whether the smaller, easier-to-train baseline models are already sufficient for this specific localization task.

During baseline testing, these models were given the same prompt and the same scanned newspaper pages. The prompt is shown in the following Figure 1.

Task: Detect the bounding box for the person labeled "{ label }" in the provided image.

Output Requirements:

Format: Return a single JSON object containing only the key "bbox_2d".

Coordinates: Use a normalized coordinate system from 0 to 1000.

Ordering: The list must follow the strict order [xmin, ymin, xmax, ymax].

Constraint: Return only the raw JSON string.

Example Output:

```
{"bbox_2d": [xmin, ymin, xmax, ymax]}
```

Figure 1: Prompt supplied to all models during the baseline experimentation phase

¹<https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct>

²<https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>

³<https://huggingface.co/Qwen/Qwen2.5-VL-32B-Instruct>

⁴<https://huggingface.co/models>

2.2 Model interface

To evaluate vision-language models, we developed an abstraction layer that provides a unified interface across them. Each model is wrapped in a class that implements a standard `predict` method. This method accepts a list of images and a text prompt, which we generate using the *Jinja2* templating engine.

The `predict` method returns a structured `Response` object, which contains the model’s raw text output, the number of tokens used, and a success flag. If an inference fails, the error message is captured and stored.

2.3 Model output

For each annotation performed by the model, the system saves a record containing the model’s output and execution metadata. These records are serialized as JSON for later analysis. Each entry follows the format described in Figure 2.

```
1 {
2     "image_path": "<uuid>/<uuid>.jpg",
3     "image_id": "<uuid>",
4     "model": "local:qwen_2_5_vl_3b_instruct:models/qwen_2_5_vl_3b_instruct",
5     "prompt_path": "prompts/annotate_prompt.j2",
6     "label": "František Josef I.",
7     "expected_bounding_box": [385, 615, 573, 796],
8     "raw_response": "...",
9     "annotation_result": [234, 725, 568, 991],
10    "success": 1,
11    "error_message": "",
12    "start_timestamp": "2026-04-28T20:10:48.853979",
13    "end_timestamp": "2026-04-28T20:10:52.852432",
14    "total_time_seconds": 3.998453,
15    "tokens_used": 35
16 }
17
```

Figure 2: Annotation result stored for each model inference

3 Realization of proposed approach

To improve the models' performance relative to their baselines, the proposed approaches of few-shot learning (section 3.1) and model finetuning via LoRA (section 3.2) are discussed.

3.1 Few-shot learning approach

The few-shot approach leverages in-context learning by providing the model with reference examples within the prompt. This strategy guides the model's reasoning by demonstrating the mapping between visual entities on a newspaper page (the target person's name label and photograph) and their corresponding bounding box coordinates.

Dynamic Template Injection The annotation process utilizes a Jinja2-based prompting engine. Instead of static text, the template contains hardcoded paths to concrete image files. During runtime, the system parses these paths, locates the associated ground-truth data, and dynamically injects the correct person labels and bounding box coordinates (`xmin`, `ymin`, `xmax`, `ymax`) into the final prompt.

Multimodal Prompting The final rendered prompt is provided to the VLM along with the physical newspaper image file. The model receives three "solved" multimodal pairs following the target query. It's example is shown in the figure 3.

Task: Detect the bounding box for the person labeled "{ label }" in the provided image.

Output Requirements:

Format: Return a single JSON object containing only the key "bbox_2d".

Coordinates: Use a normalized coordinate system from 0 to 1000.

Ordering: The list must follow the strict order [xmin, ymin, xmax, ymax].

Constraint: Return only the raw JSON string.

Example 1:
Image: <uuid_1>/<uuid_1>.jpg
Label: {{ label_1 }}
Output:
{{ bbox_2d_1 }}

Example 2:
Image: <uuid_2>/<uuid_2>.jpg
Label: {{ label_2 }}
Output:
{{ bbox_2d_2 }}

Example 3:
Image: <uuid_3>/<uuid_3>.jpg
Label: {{ label_3 }}
Output:
{{ bbox_2d_3 }}

Figure 3: Prompt supplied to all models during the few-shot experimentation phase, with three concrete examples of newspaper scans with the target person name (label) and the desired bounding box coordinates

3.2 Finetuning approach using LoRA

Low-Rank Adaptation (LoRA) is utilized to efficiently adapt the `Qwen2.5-VL` models by injecting trainable low-rank matrices into the Transformer architecture while keeping the backbone weights frozen. This approach is essential for large model fine-tuning as it reduces VRAM requirements and prevents "catastrophic forgetting," ensuring that the model retains its foundational visual-linguistic reasoning while learning new task-specific patterns.

Rank ($r = 16$) The rank determines the dimensionality of the update matrices. A value of 16 was selected for the `Qwen-3B` to provide sufficient learning power for complex vision-language tasks. While lower ranks (e.g., 4 or 8) further reduce the number of parameters, a rank of 16 ensures the model can capture the nuanced spatial and textual relationships required for precise bounding box generation and image description.

Targeted Layers Fine-tuning targets all matrices within attention and feed-forward blocks (MLP) specifically the key, value, query matrices and the output projection (`o_proj`), along with the gate (`gate_proj`), up (`up_proj`), and down (`down_proj`) matrices of the MLP.

4 Experiments

In the upcoming subsections, multiple experiments are proposed and evaluated. Firstly, the baseline experiment, where the three base-selected models’ performances are compared. Secondly, the few-shot learning performance of the Qwen-3B model is compared to its baseline. Finally, the finetuning via the LoRA approach is evaluated, and the finetuned Qwen-3B model’s performance is compared to its baseline.

4.1 Baseline experiment

In this initial phase, we conducted a series of baseline experiments to evaluate the performance of various models on our specific dataset. The primary task for each model was to localize the bounding box of a given person within the newspaper page. The number of testing samples was 311.

F1 score comparison

To objectively evaluate the models, we used the F1 score across various Intersection over Union (IoU) thresholds. The results of this comparison are summarized in Table 1 below:

Model Name	Avg IoU	F1@.10	F1@.20	F1@.30	F1@.40	F1@.50	F1@.60	Validity
Qwen-3B	0.0642	0.2793	0.0897	0.0345	0.0103	0.0034	0.0034	86%
Qwen-7B	0.0688	0.3080	0.0920	0.0460	0.0248	0.0035	0.0035	82%
Qwen-32B	0.0611	0.2610	0.0983	0.0373	0.0068	0.0034	0.0034	90%

Table 1: Expanded model comparison with average IoU, F1 across thresholds, and successful predictions (valid bounding box generated). Zero F1 scores for higher thresholds were omitted due to clarity. Highlighted cells show the best value in a column.

To more clearly illustrate the performance decay as spatial precision requirements increase, the data from Table 1 is visualized in Figure 4.

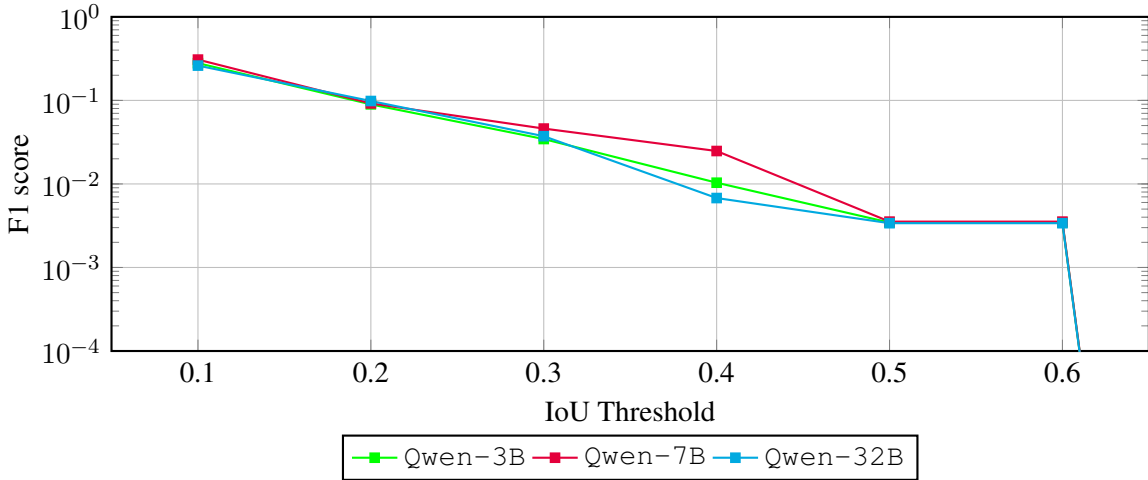


Figure 4: F1 scores of baseline models across IoU thresholds. Zero F1 scores for higher thresholds were omitted due to clarity.

Conclusion

The baseline experiments reveal that increasing model scale does not guarantee a proportional improvement in zero-shot localization. While the Qwen-7B model achieved the highest Average IoU (0.0688), the Qwen-32B

model generated the highest amount of valid bounding boxes (90%), despite having the lowest Average IoU (0.0611). This suggests that larger models may be more reliable at adhering to output formatting constraints.

The sharp drop-off in F1 scores beyond the 0.2 IoU threshold for all models underscores the difficulty of the newspaper dataset. None of the general-purpose models demonstrates sufficient precision for practical localization in this domain without further adaptation.

4.2 Few-shot experiment

Following the zero-shot baseline evaluations, we applied a few-shot prompting strategy to the Qwen-3B model. By providing the model with a limited set of labeled examples within the prompt context, we aimed to assess whether in-context learning could mitigate the challenges of person localization in old newspaper scans without requiring weight updates.

F1 Score Comparison

The evaluation focuses on the F1 score across various IoU thresholds to measure localization accuracy. Table 2 compares the performance of the zero-shot baseline against the few-shot approach.

Model Name	Avg IoU	F1@.10	F1@.20	F1@.30	F1@.40	F1@.50	F1@.60	F1@.70	Validity
Qwen-3B (Base)	0.0642	0.2793	0.0897	0.0345	0.0103	0.0034	0.0034	0.0000	86.49%
Qwen-3B (Few-shot)	0.0414	0.1298	0.0599	0.0399	0.0200	0.0067	0.0033	0.0033	93.25%

Table 2: Comparison between the baseline Qwen-3B model and the few-shot prompted version. Interestingly, while the few-shot method improved the success rate, the average IoU actually decreased compared to the zero-shot baseline.

The performance divergence is visualized in Figure 5. On a logarithmic scale, it becomes evident that while few-shot prompting helps the model maintain a marginal presence at higher IoU thresholds (up to 0.7), it suffers from a significant drop in precision at lower thresholds compared to the zero-shot baseline.

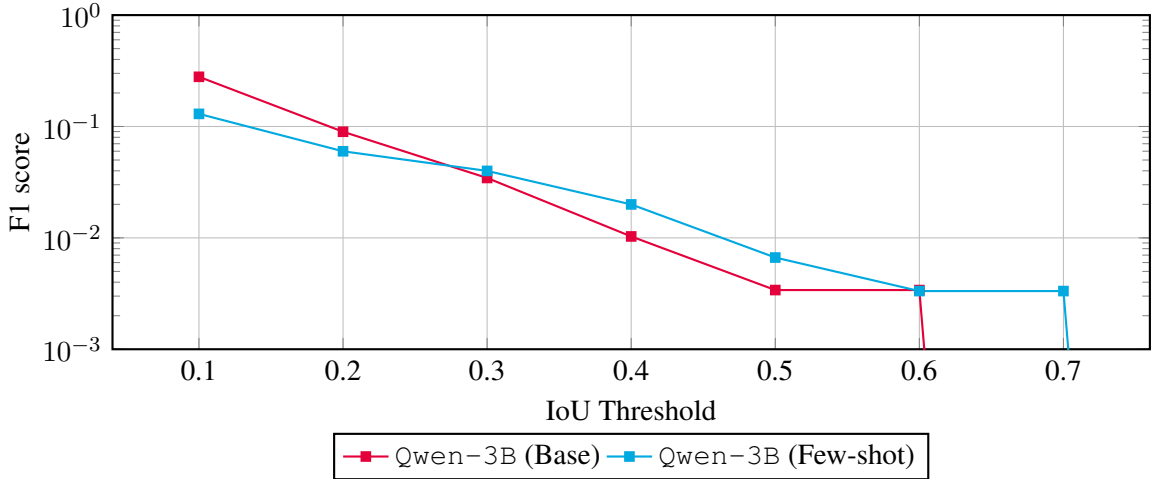


Figure 5: F1 scores of baseline and few-shot 3B model across the range of IoU thresholds.

Output Consistency and Format Adherence

A notable benefit of the few-shot approach is the improvement in structural reliability. By providing examples of the expected JSON or coordinate format in the prompt, the model’s “Validity” rate (the ability to produce a syntactically valid bounding box) increased from 86.49% (269/311) to 93.25% (290/311).

This reduction in formatting errors indicates that the Qwen-3B model can mimic the output structure of the provided examples. However, this “operational reliability” did not translate into high “spatial accuracy.” While the model produced 21 fewer malformed responses than the baseline, its bounding boxes were often farther from the ground truth, as evidenced by a drop in Average IoU from 0.0642 to 0.0414.

Performance Multiplier Analysis

Despite the lower overall F1 score at low thresholds, the few-shot model exhibits a different performance profile than the baseline as constraints tighten. By analyzing the performance multiplier, we can identify where the examples most improved the model’s performance.

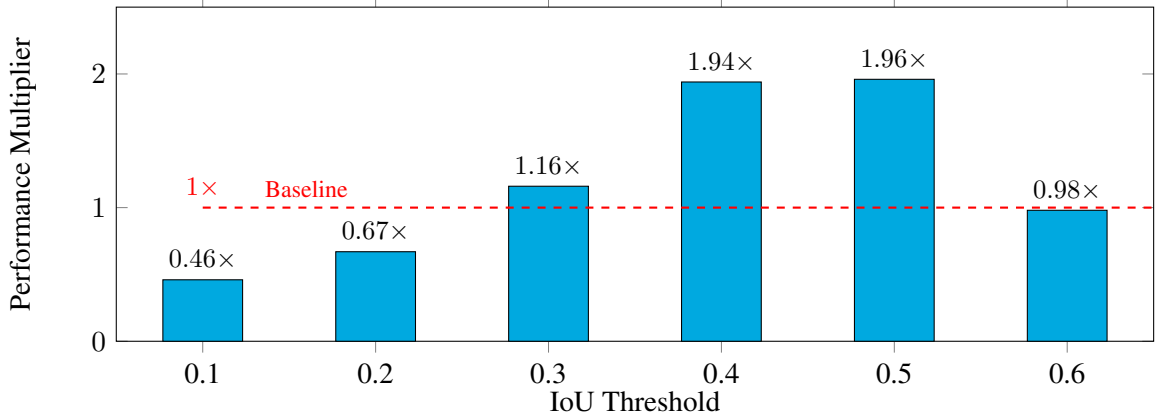


Figure 6: Performance multiplier of the Few-shot Qwen-3B model relative to the Baseline. Values below $1\times$ indicate a regression in performance compared to the zero-shot approach.

As shown in Figure 6, the few-shot model initially underperforms the baseline ($0.46\times$ at IoU 0.1). However, at the 0.4 and 0.5 IoU thresholds, the model becomes nearly twice as effective as the baseline.

Conclusion

The comparison between zero-shot and few-shot methods reveals a trade-off. The few-shot strategy effectively improved the model’s adherence to output constraints (increasing the validity rate to 93.25%), but it resulted in a regression in overall spatial localization, with Average IoU falling by approximately 35.51%.

4.3 Finetuning experiment

Lastly, this experiment aims to determine how effectively a smaller-scale Qwen-3B model can be optimized using Low-Rank Adaptation (LoRA).

F1 score comparison

The comparative results between the zero-shot baseline and our finetuned version are presented in Table 3.

Model Name	Avg IoU	F1@.10	F1@.20	F1@.30	F1@.40	F1@.50	F1@.60	F1@.70	F1@.80	Validity
Qwen-3B (Base)	0.0642	0.2793	0.0897	0.0345	0.0103	0.0034	0.0034	0.0000	0.0000	86.50%
Qwen-3B (LoRA)	0.2691	0.6634	0.5475	0.4090	0.3027	0.2222	0.1224	0.0322	0.0129	99.68%

Table 3: Comparison between the baseline Qwen-3B model and the LoRA finetuned version. The finetuned model shows a significant improvement in both spatial accuracy (Avg IoU) and overall localization performance across all thresholds.

Visualization of the Table 3 can be viewed in Figure 7. This highlights the finetuned model’s ability to maintain localization consistency even as the IoU requirements become more strict, a scenario where the baseline model’s performance essentially collapses.

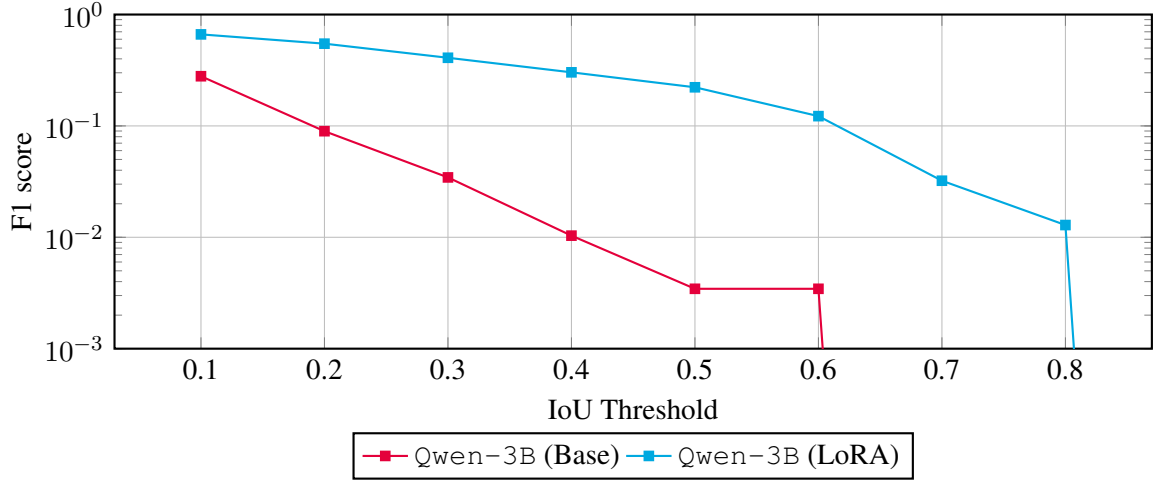


Figure 7: F1 scores of baseline and finetuned 3B model across the range of IoU thresholds.

Output Consistency and Format Adherence

While the baseline experiments showed that the zero-shot model often struggled to produce coordinates within the valid range of the document’s dimensions, the LoRA-finetuned *Qwen-3B* model demonstrated a near-total elimination of these structural errors. This indicates that the training process effectively improved the model’s ability to adhere to the requested format, ensuring that nearly every response was a syntactically correct and spatially plausible bounding box.

Performance improvement

By calculating the performance multiplier at each threshold, we can see the specific precision levels at which LoRA adaptation provides the greatest benefit.

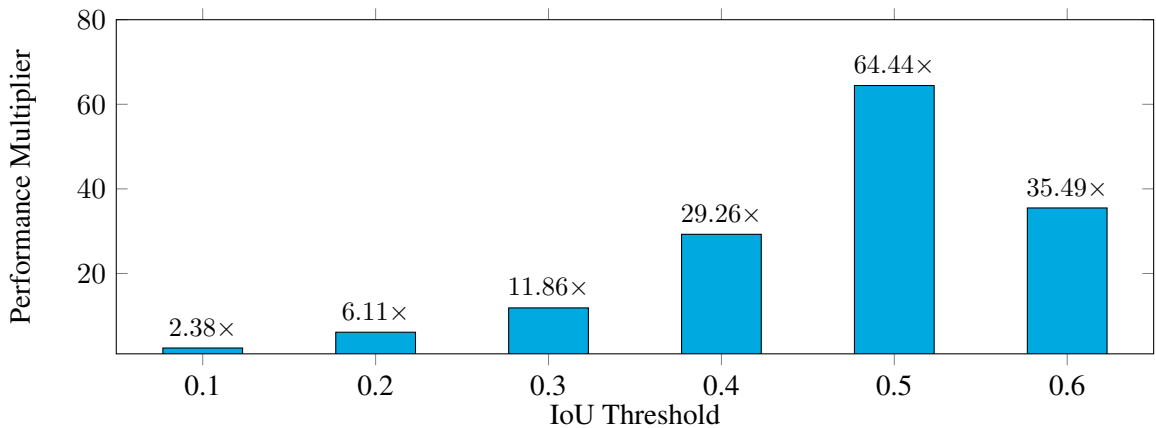


Figure 8: Performance multiplier of the Finetuned *Qwen-3B* model relative to the Baseline across IoU thresholds.

As shown in Figure 8, the finetuned model demonstrates a massive increase in reliability. The most substantial leap occurs at the 0.5 IoU threshold, where the finetuned model is over 64 times more accurate than the

baseline. This indicates that while the baseline model occasionally "guesses" the general position of a person's portrait, the finetuned model has learned the structural patterns and traits required for better localization.

Conclusion

The transition from 0.0642 to 0.2691 in Average IoU confirms that targeted finetuning is essential for complex document analysis. A notable finding is the improvement in the model's structural output: the finetuned model achieved a near-perfect success rate (99%) for "Validity" meaning it almost always generated a syntactically valid bounding box within the page boundaries.

While a valid bounding box does not inherently guarantee the person was correctly located, this high success rate represents a significant reduction in hallucinated or out-of-bounds coordinates compared to the baseline. Coupled with the improved F1 scores, these results prove that model size is less of a bottleneck than domain-specific training. Even a small model, when properly tuned, can learn a task's spatial constraints far better than much larger general-purpose models.

5 Conclusion

This work investigated whether modern vision-language models (VLMs) can localize portrait photographs of named individuals in scanned historical newspapers, a domain characterized by complex layouts, low contrast, and significant degradation. Using the PeopleGator dataset (~2,500 annotated pages), we compared zero-shot prompting, few-shot in-context learning, and domain adaptation via LoRA fine-tuning on models from the Qwen2.5-VL family.

Across the zero-shot baseline, scaling model size from 3B to 7B and 32B did not yield meaningful gains in localization accuracy (average IoU around 0.06–0.07, with a steep F1 drop beyond IoU 0.2). Few-shot prompting improved output robustness (valid bounding boxes increased from 86.49% to 93.25%), but did not reliably improve spatial precision, and the average IoU decreased from 0.0642 to 0.0414. In contrast, LoRA fine-tuning of the Qwen-3B model produced a decisive improvement: average IoU increased to 0.2691, validity reached 99.68%, and F1 scores improved consistently across all IoU thresholds (e.g., F1@0.5 improved approximately 64 times). These results show that domain-specific training is far more impactful than model scaling or prompt-only adaptation for historical newspaper localization.

References

- [1] Edward J. Hu et al. *LoRA: Low-Rank Adaptation of Large Language Models*. 2021. arXiv: 2106.09685 [cs.CL]. URL: <https://arxiv.org/abs/2106.09685>.
- [2] Marek Vasko, Adam Herout, and Michal Hradis. “Archival Faces: Detection of Faces in Digitized Historical Documents”. In: *CoRR* abs/2504.00558 (2025). DOI: 10.48550/ARXIV.2504.00558. arXiv: 2504.00558. URL: <https://doi.org/10.48550/arXiv.2504.00558>.